

THE STRUCTURE OF MATHEMATICAL REALITY

Phase I — Chapter 2

What Numbers Actually Are

Note Template

Self-Study Curriculum

How to use this template

Complete these notes **after** the reading and the lecture. The goal is not to copy phrases from the text — it is to reconstruct the ideas in your own language. Leave every item blank on first pass; fill it in from memory, then check.

1. Peano's axioms define the natural numbers by

list the key structural requirements in your own words

2. The principle: an object is defined by its _____, not its _____.

3. The chain of number system extensions

$\mathbb{N} \rightarrow \mathbb{Z}$: forced by the need to close under _____

$\mathbb{Z} \rightarrow \mathbb{Q}$: forced by the need to close under _____

$\mathbb{Q} \rightarrow \mathbb{R}$: forced by the need to _____

$\mathbb{R} \rightarrow \mathbb{C}$: forced by the need to _____

4. What is lost at each step

$\mathbb{Z}$ loses:

$\mathbb{Q}$ loses:

$\mathbb{R}$ loses:

$\mathbb{C}$ loses:

5. Why the number line is a partial picture

It shows: _____ and _____

It hides: _____ and _____

6. The connection between complex numbers and rotation

explain in your own words why multiplication by a complex number is a rotation

7. What “algebraically closed” means and why it matters

your words

8. Countable vs. uncountable

Two infinite sets have “the same size” when:

A set is countable when:

The diagonal argument shows:

The same diagonal maneuver will return in:

Chapter _____ (as _____) and Chapter _____ (as _____)

9. The thing I previously assumed about numbers that I now see

differently:

one specific assumption
